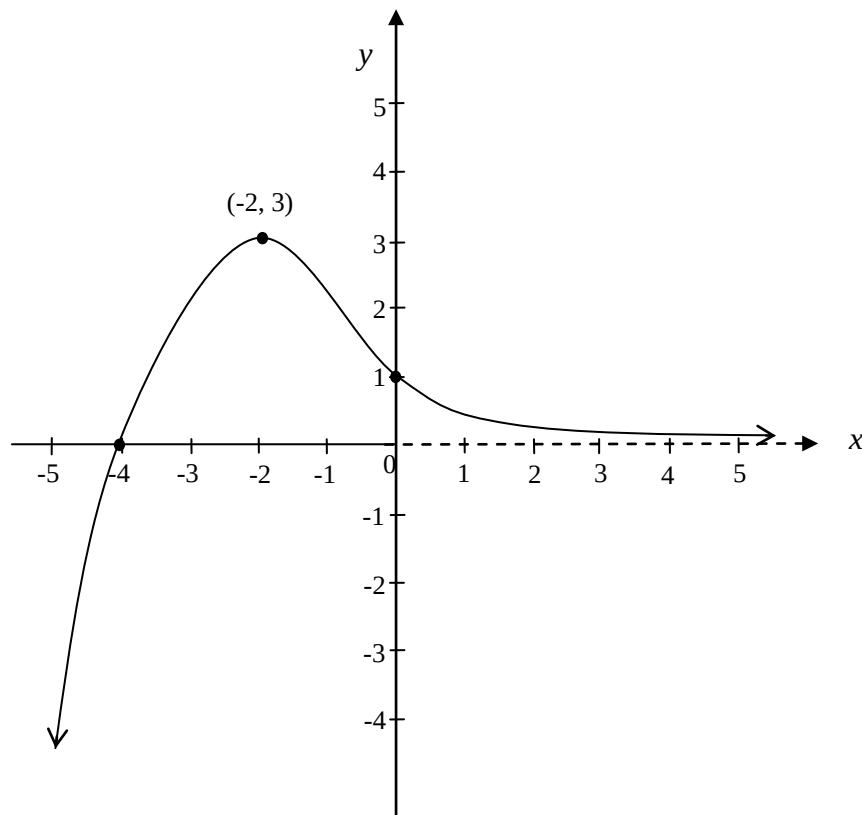


QUESTION 1

- | | | | |
|-----|--|---|---------------------------------|
| (a) | Evaluate: | $\int_0^4 \frac{x}{\sqrt{9+x^2}} dx.$ | Marks
2 |
| (b) | Find: | <p>(i) $\int \frac{1}{1+e^{-x}} dx.$</p> <p>(ii) $\int \sec x \tan^3 x dx.$</p> | <p>2</p> <p>2</p> |
| (c) | An ellipse has the equation $\frac{x^2}{4} + y^2 = 1.$ | <p>(i) Calculate the eccentricity for this ellipse.</p> <p>(ii) Draw a neat sketch of the ellipse, clearly labelling the foci, directrices and intercepts with the coordinate axes.</p> | <p>2</p> <p>3</p> |
| (d) | Find the coordinates of the points on the graph of $x^3 + y^3 = 3xy$ at which the tangent lines are parallel to the x -axis. | | 4 |

QUESTION 2 START A NEW PAGE

- (a) The diagram below shows the graph of $y = f(x)$. The graph has a horizontal asymptote at $y = 0$.



Draw, on separate sets of axes, sketches of the following graphs.

- | | | | |
|------|----------------|--|----------|
| (i) | $y = f(x)$ | | 1 |
| (ii) | $y = 2^{f(x)}$ | | 3 |

Question 2 continued

(b) Given $I_n = \int_0^1 \frac{x^n}{x^2 + 1} dx$ for $n = 1, 2, 3, \dots$ **Marks**

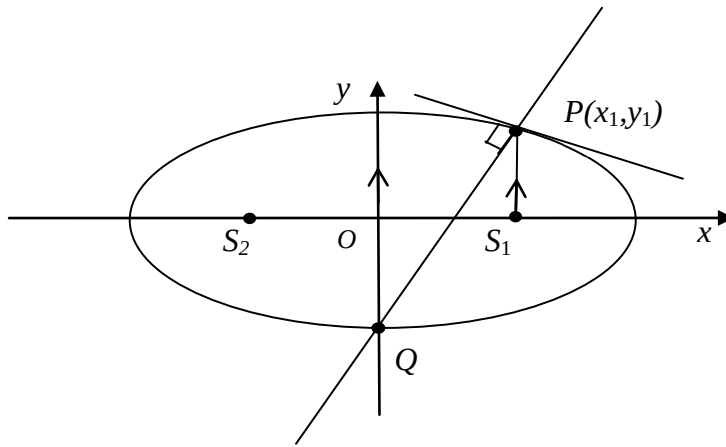
(i) Show that:

$$I_n = \frac{1}{n-1} - I_{n-2} \text{ for } n \geq 2. \quad 2$$

(ii) Hence, evaluate:

$$\int_0^1 \frac{x^5}{x^2 + 1} dx. \quad 2$$

(c) $P(x_1, y_1)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci S_1 and S_2 , so that PS_1 is parallel to the y -axis and $y_1 \geq 0$, as shown in the diagram below.



(i) Show that the y -coordinate of P can be given by $y_1 = a(1 - e^2)$, where e is the eccentricity of the ellipse. **2**

(ii) Prove that the equation of the normal at P is $x - ey - ae^3 = 0$. **3**

(iii) For a particular ellipse the normal at P passes through point Q which is at the end of the minor axis, as shown.

Calculate the value of e^2 for this ellipse, expressing your answer as a surd in simplest form. **2**

QUESTION 3 START A NEW PAGE

Marks

- (a) A hyperbola has foci at $S_1(0, 6)$ and $S_2(0, -6)$. One of the vertices is at $A(0, -2)$.
- (i) Find the equation of the hyperbola. **3**
- (ii) Find the equations of the directrices and the asymptotes of the hyperbola. **2**
- (b) Consider the function $f(x) = e^{-x} \sin x$.
- (i) Show that the graph of $y = f(x)$
- (a) intersects the x -axis at $x = n\pi$, where n is an integer, **1**
- (b) has stationary points at $x = \frac{(4n+1)\pi}{4}$. **1**
- (ii) Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$. **2**
- (iii) Show that $\int e^{-x} \sin x \, dx = -\frac{1}{2} e^{-x} (\sin x + \cos x) + c$, where c is a constant. **3**
- (iv) If A_n is the magnitude of the area of the region bounded by the curve $y = e^{-x} \sin x$ and the x -axis for $(n-1)\pi \leq x \leq n\pi$ show that: **3**
- $$\frac{A_1}{A_0} = e^{-\pi}.$$

QUESTION 4 START A NEW PAGE

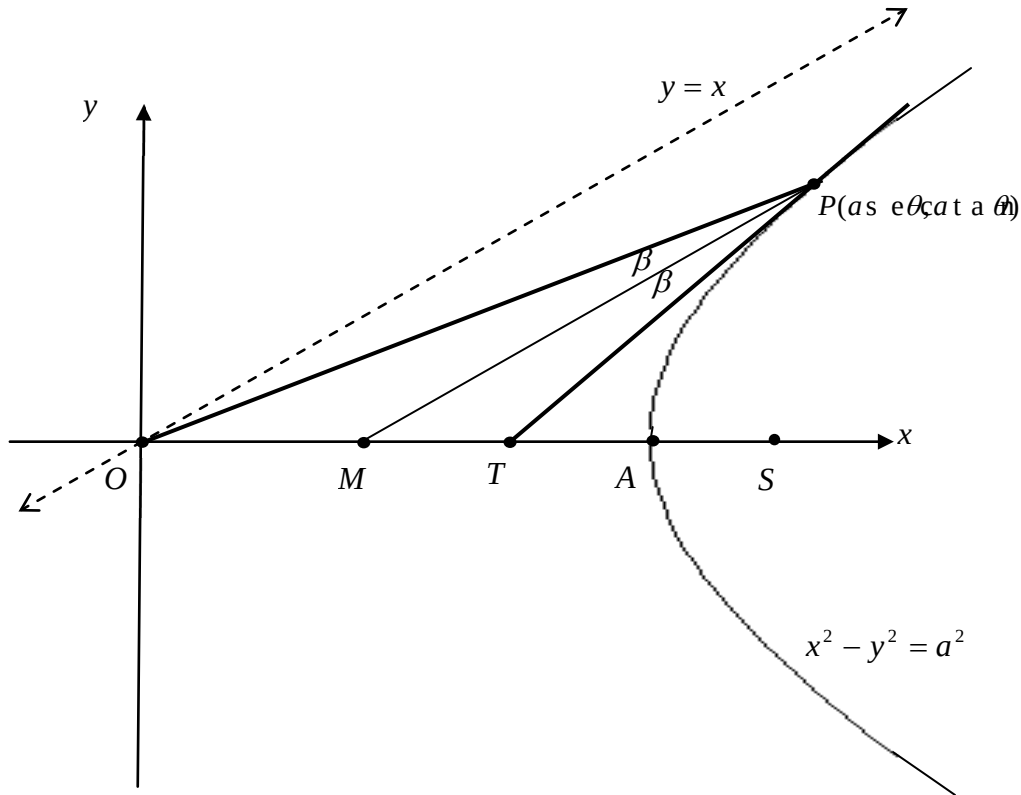
Marks

- (a) (i) Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. **2**
- (ii) Hence, or otherwise, calculate the value of $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$. **4**

Question 4 continued

Marks

- (b) $P(a \sec \theta, a \tan \theta)$ is a point on the hyperbola $x^2 - y^2 = a^2$, where $0 < \theta < \frac{\pi}{2}$ as shown in the diagram.



The point of intersection of the tangent at P with the x -axis is point T . O is the origin.
Let $\angle OPT = 2\beta$,

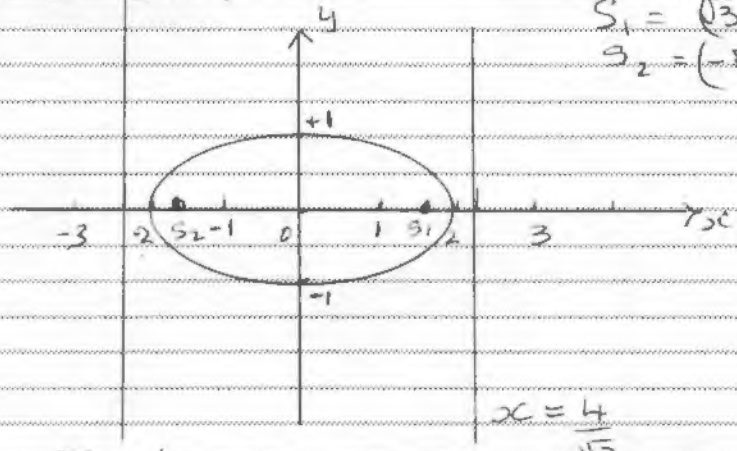
- (i) Show that $\tan 2\beta = \frac{\cos^2 \theta}{2 \sin \theta}$. **3**
- (ii) By using the formula $\tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta}$, show that $\tan \beta = \left[\frac{1 - \sin \theta}{\cos \theta} \right]^2$. **3**
- (iii) Given that MP is the bisector of $\angle OPT$, as shown in the diagram above, prove that MP is always parallel to one asymptote of the hyperbola. **3**

END OF EXAMINATION

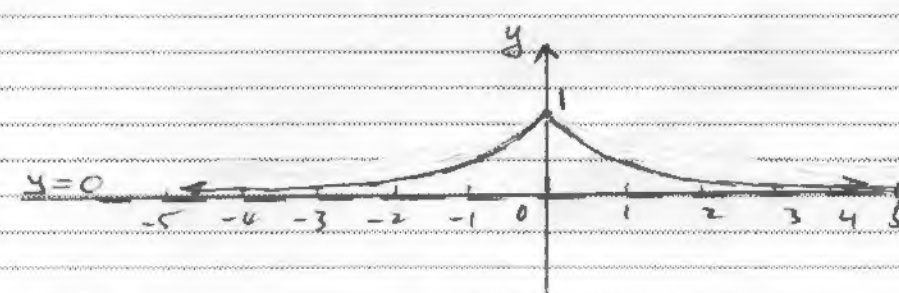
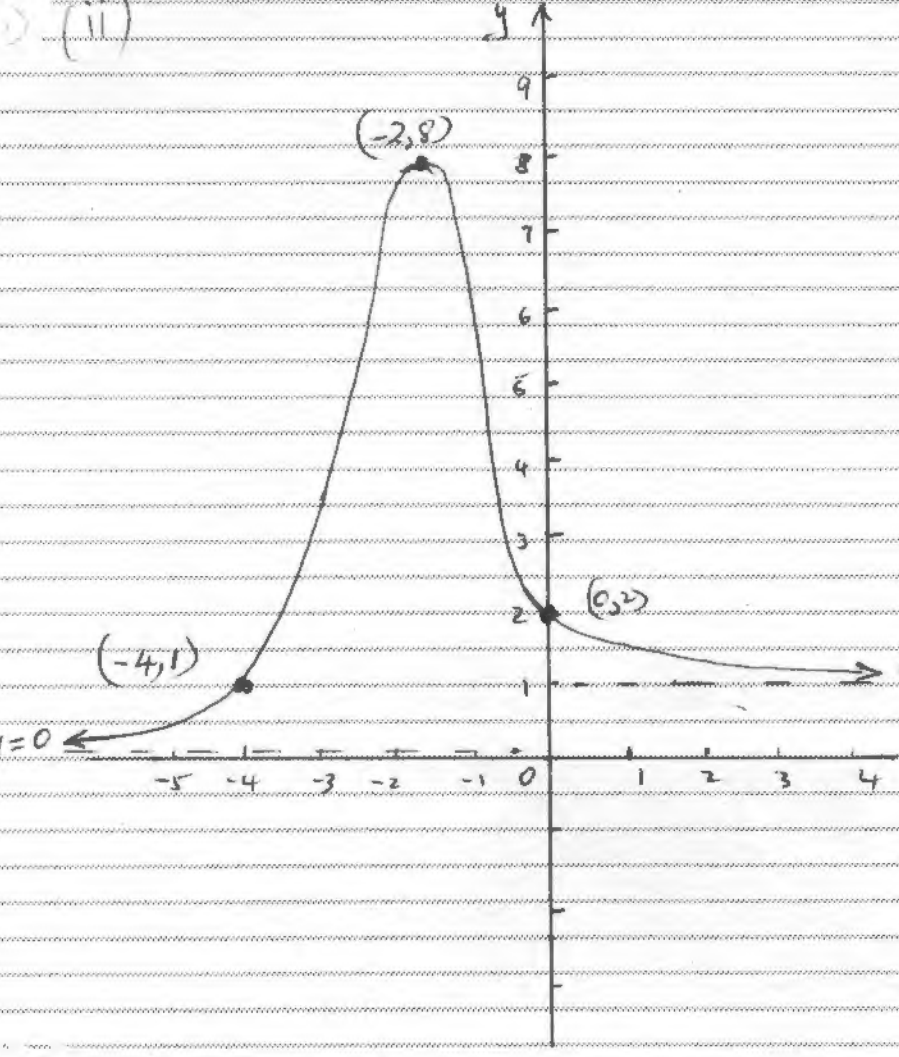
MATHEMATICS Extension 2: Question 1

Suggested Solutions	Marks	Marker's Comments
a) $\int_0^4 \frac{x}{\sqrt{9+x^2}} dx$ $= \frac{1}{2} \int_0^4 2x (9+x^2)^{-\frac{1}{2}} dx$ $= \frac{1}{2} \left[\frac{(9+x^2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_0^4$ $= \left[\sqrt{9+x^2} \right]_0^4 = \sqrt{9+16} - \sqrt{9+0}$ $= 5 - 3 = 2$	(2)	① Integration ① Answer
b) (i) $\int \frac{1}{1+e^{-x}} dx$ $= \int \frac{1}{1+e^{-x}} \times \frac{e^x}{e^x} dx$ $= \int \frac{e^x}{e^x+1} dx$ $= \ln(e^x+1) + C$	(2)	① change of integral ① answer
(ii) $\int \sec x \tan^3 x dx$ $= \int \sec x (\tan^2 x) \tan x dx$ $= \int \sec x (\sec^2 x - 1) \tan x dx$ $= \int \sec^3 x \tan x - \sec x \tan x dx$ $= \int \sec x \tan x (\sec^2 x) - \sec x \tan x dx$ $= \frac{\sec^3 x}{3} - \sec x + C$	(2)	① use of identity and re organisation of integral ① answer
c) i) $\frac{x^2}{4} + y^2 = 1 \quad a=2 \quad y=1$ $b^2 = a^2 (1 - e^2)$ $1 = 4 (1 - e^2)$ $\frac{1}{4} = 1 - e^2$ $e^2 = 1 - \frac{1}{4} = \frac{3}{4} \quad \therefore e = \frac{\sqrt{3}}{2}$	(2)	correct sub into formula ① answer

MATHEMATICS Extension 2: Question 1

Suggested Solutions	Marks	Marker's Comments
c) (ii) Foci: $2c = \pm ae$ $y = 0$ $ae = 2 \times \sqrt{3} = \sqrt{3} \sim 1.7$ Directrices $x = \pm \frac{a}{e} = \frac{2}{\frac{\sqrt{3}}{2}} = \pm \frac{4}{\sqrt{3}} \sim \pm 2.3$ $a = 2$ $b = 1$  $S_1 = (3, 0)$ $S_2 = (-3, 0)$ $x = \frac{4}{\sqrt{3}}$ $x = -\frac{4}{\sqrt{3}}$	③	① foci ① directrices ① scale & shape, axes, etc
d) $x^3 + y^3 = 3xy$ (i) $3x^2 + 3y^2 \frac{dy}{dx} = 3x \frac{dy}{dx} + 3y$ for horizontal tangent $\frac{dy}{dx} = 0$ $\therefore y = x^2$ sub into (i) $x^3 + (x^2)^3 = 3x(x^2)$ $\therefore x^3 + x^6 = 3x^3$ $\therefore x^6 - 2x^3 = 0$ $x^3(x^3 - 2) = 0$ $\therefore x = 0$ or $x = \sqrt[3]{2}$ $y = 0$ $y = \sqrt[3]{4}$ Points $(0, 0)$, $(\sqrt[3]{2}, \sqrt[3]{4})$	④	① differentiation ① $y = x^2$ ① x values ① y values. (or ① for each point)

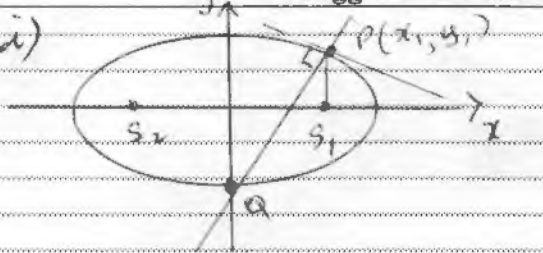
MATHEMATICS Extension 2: Question 2

Suggested Solutions	Marks	Marker's Comments
(a) (i) 	①	must show asymptote $y=0$ reflection about $x=0$
(ii) 	③	① Asymptotes $y=0$ $y=1$ ① y intercept $(0,2)$ Point $(4,1)$ ① Max at $(-2,8)$ & shape

MATHEMATICS Extension 2: Question 2...continued

Suggested Solutions	Marks	Marker's Comments
b) (i) $I_n = \int_0^1 \frac{x^n}{x^2+1} dx \quad n=1,2,3,\dots$ $= \int_0^1 x^{n-2} \times \frac{x^2}{(x^2+1)} dx$ $= \int_0^1 x^{n-2} \times \left[1 - \frac{1}{x^2+1} \right] dx$ $= \int_0^1 x^{n-2} dx - \int_0^1 \frac{x^{n-2}}{x^2+1} dx$ $= \left[\frac{x^{n-1}}{n-1} \right]_0^1 - I_{n-2}$ $= \frac{1}{n-1} - I_{n-2}$	(2)	① rearranging integral ① integration
$\int_0^1 \frac{x^5}{x^2+1} dx = I_5$ $I_5 = \frac{1}{5-1} - I_3$ $= \frac{1}{4} - \left[\frac{1}{3-1} - I_1 \right]$ $= \frac{1}{4} - \left[\frac{1}{2} - \int_0^1 \frac{x}{x^2+1} dx \right]$ $= \frac{1}{4} - \frac{1}{2} + \left[\frac{1}{2} \ln(x+1) \right]_0^1$ $= -\frac{1}{4} + \frac{1}{2} \ln 2 - \frac{1}{2} \ln 1$ $= \frac{1}{2} \ln 2 - \frac{1}{4}$	(2)	correct sub to I_1 ① answer

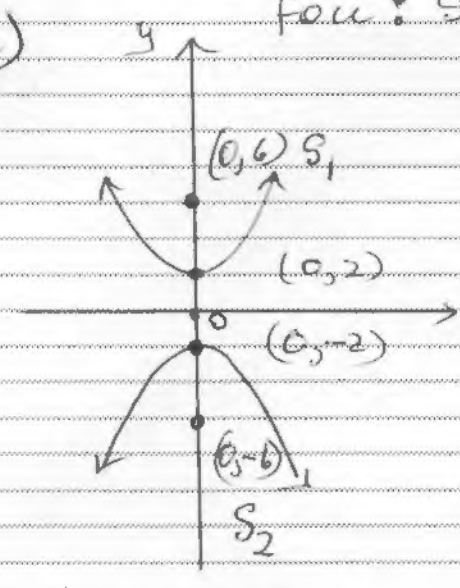
MATHEMATICS Extension 2: Question 2 (continued)

Suggested Solutions	Marks	Marker's Comments
(i)  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $x_1 = ae$ $b^2 = a^2(1-e^2)$ $\frac{(ae)^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1$ $e^2 + \frac{y^2}{a^2(1-e^2)} = 1$ $y^2 = (1-e^2) \times a^2(1-e^2)$ $y = a(1-e^2) \text{ as } y_1 \geq 0$	(2)	{1} ① showing result.
(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{2ac}{a^2} \times \frac{b^2}{2y}$ $x = ae \quad y = a(1-e^2)$ $m_T = -\frac{2(ae)}{a^2} \times \frac{a^2(1-e^2)}{2a(1-e^2)}$ $= -e$ $m_N = \frac{1}{e}$ Equation of normal: $y - y_1 = m(x - x_1)$ $y - a(1-e^2) = \frac{1}{e}(x - ae)$ $y - a + ae^2 = \frac{x}{e} - a$ $y + ae^2 = \frac{x}{e}$ $0 = x - ey - ae^3$ Normal is $x - ey - ae^3 = 0$	(3)	① $\frac{dy}{dx}$ formula ① grad of tangent. ① equation

MATHEMATICS Extension 2: Question... 2 continued

Suggested Solutions	Marks	Marker's Comments
(c)(iii) $Q = (0, -b)$ $b^2 = a^2(1-e^2)$ $\therefore x=0$ $y = -a\sqrt{1-e^2}$ $x - ey - ae^3 = 0$ $+ea\sqrt{1-e^2} = ae^3$ $\sqrt{1-e^2} = e^2$ $1-e^2 = e^4$ $e^4 + e^2 - 1 = 0$ $\therefore e^2 = \frac{-1 \pm \sqrt{1+4}}{2}$ but $e^2 > 0$ $\therefore e^2 = \frac{-1 + \sqrt{5}}{2}$	(2)	an equation in e only ① answer

MATHEMATICS Extension 2: Question 3

Suggested Solutions	Marks	Marker's Comments
Q1)  foci: $S_1 = (0, 6)$ $S_2 = (0, -6)$ $\therefore$ centre is $(0, 0)$ Vertex = $(0, 2)$ $\therefore$ Hyperbola is of the form: $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ $b = 2$ $be = 6$ $2e = 6$ $e = 3$ $a^2 = b^2(e^2 - 1)$ $= 4(9 - 1)$ $= 32$ Equation of Hyperbola is $\frac{y^2}{4} - \frac{x^2}{32} = 1$	(3)	① basic equation of hyperbola ① value of e ① answer
(ii) Directrices $y = \pm \frac{b}{e} \quad y = \pm \frac{2}{3}$ Asymptotes $x = \pm \frac{a}{b} y$ $x = \pm \frac{\sqrt{32}}{2} y$ $x = \pm 2\sqrt{2} y$ OR $y = \pm \frac{1}{2\sqrt{2}} x$	(2)	① answer ① answer

MATHEMATICS Extension 2: Question 3

Suggested Solutions	Marks	Marker's Comments
b) i) $y = e^{-x} \sin x$ 2) x intercepts $y = 0$ $0 = e^{-x} \sin x \quad e^{-x} \neq 0$ $\therefore \sin x = 0$ $x = n\pi \quad n \in \mathbb{Z}$	①	must include $e^{-x} \neq 0$ ①
(b) stationary points $\frac{dy}{dx} = 0$ $\frac{dy}{dx} = e^{-x} \cos x - e^{-x} \sin x$ $0 = e^{-x} (\cos x - \sin x)$ $\cos x = \sin x \quad e^{-x} \neq 0$ $\tan x = 1$ $x = n\pi + \frac{\pi}{4}$ $x = (4n+1)\frac{\pi}{4}$	①	no loss of marks if not write $e^{-x} \neq 0$ ①
(ii)	②	Accurate graph not required. ① Must show • intercepts at $x = \pm\pi$ and 0 • turning points at $x = -3\pi/4$ and $x = \pi/4$ Approx values are: (0.8, 0.3) (2.4, -7.5)

MATHEMATICS Extension 2: Question 3

Suggested Solutions	Marks	Marker's Comments
(iii) $I = \int e^{-x} \sin x \, dx =$ $= [-e^{-x} \sin x] + \int e^{-x} \cos x \, dx$ $= [-e^{-x} \sin x] + [-e^{-x} \cos x - \int e^{-x} \sin x \, dx]$ $= -e^{-x} (\sin x + \cos x) - \int e^{-x} \sin x \, dx$ $= -e^{-x} (\sin x + \cos x) - I + C$ $\therefore 2I = -e^{-x} (\sin x + \cos x) + C$ $I = -\frac{1}{2} e^{-x} (\sin x + \cos x) + C$ (iv) $A_1 = \left[-\frac{1}{2} e^{-x} (\sin x + \cos x) \right]_0^\pi$ $= \left\{ -\frac{1}{2} e^{-\pi} (0 - 1) \right\} - \left\{ -\frac{1}{2} e^0 (0 + 1) \right\}$ $= \frac{1}{2} e^{-\pi} + \frac{1}{2}$ $A_0 = \left \left[-\frac{1}{2} e^{-x} (\sin x + \cos x) \right]_{-\pi}^0 \right $ $= \left \left\{ -\frac{1}{2} e^0 (0 + 1) \right\} - \left\{ -\frac{1}{2} e^{\pi} (0 - 1) \right\} \right $ $= \left -\frac{1}{2} - \frac{1}{2} e^{\pi} \right = \frac{1}{2} + \frac{1}{2} e^{\pi}$ $\frac{A_1}{A_0} = \frac{\frac{1}{2} (e^{-\pi} + 1)}{\frac{1}{2} (1 + e^{\pi})} = \frac{e^{-\pi} (1 + e^{\pi})}{(1 + e^{\pi})}$ $= e^{-\pi}$	③ ③	① first integration by parts ① 2nd integration by parts ① showing answer ① No loss of mark here for not using absolute ① ① loss of mark if absolute not used for A_0

MATHEMATICS Extension 2: Question 4

Suggested Solutions	Marks	Marker's Comments
Q) (i) LHS = $\int_0^a f(x) dx = - \int_a^0 f(a-u) du$ $u = a-x$ $= \int_0^a f(a-u) du$ $x=0 \quad u=a$ $x=a \quad u=0$ $= \text{RHS}$	(2)	① ① change of variable values.
(ii) $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \int_0^\pi \frac{(\pi-x) \sin(\pi-x)}{1 + \cos^2(\pi-x)} dx$ $= \int_0^\pi \frac{(\pi-x) \sin x}{1 + \cos^2 x} dx$ $= \int_0^\pi \frac{\pi \sin x}{1 + \cos^2 x} dx - \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$ $\therefore 2 \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx$ use substitution $y = \cos x$ $\frac{dy}{dx} = -\sin x$ $x=0 \quad y=1$ $x=\pi \quad y=-1$ $\therefore 2I = \pi \int_1^{-1} \frac{-1}{1+y^2} dy$ $\therefore I = -\frac{\pi}{2} [\tan^{-1} y]_1^{-1}$ $= -\frac{\pi}{2} [\tan^{-1}(-1) - \tan^{-1}(1)]$ $= -\frac{\pi}{2} \times [-\frac{\pi}{4} - \frac{\pi}{4}] = \frac{\pi^2}{4}$	(4)	change variable and split integral. ① and split integral. ① Move part integral to LHS ① substitution ① answer

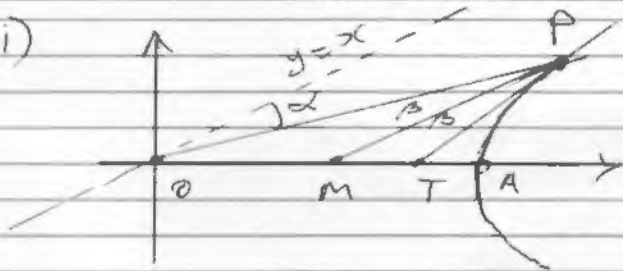
MATHEMATICS Extension 2: Question 4 continued

Suggested Solutions	Marks	Marker's Comments
b) (i) $x^2 - y^2 = a^2$ $2x - 2y \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{-2x}{-2y}$ $= \frac{x}{y}$ gradient of PT = $\frac{a \sec \theta}{a \tan \theta}$ $= \frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} = \frac{1}{\sin \theta}$ gradient of OP = $\frac{a \tan \theta}{a \sec \theta} = \sin \theta$ Angle between 2 lines is acute $\tan 2\beta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $= \frac{1 - \sin \theta}{\sin \theta \left(1 + \frac{1}{\sin \theta} \times \sin \theta \right)}$ $= \frac{1 - \sin^2 \theta}{\sin \theta [1 + 1]}$ $= \frac{\cos^2 \theta}{2 \sin \theta}$	(3)	① grad PT ① grad OP ① use of angle between two lines formula.

MATHEMATICS Extension 2: Question 4 continued

b)	Suggested Solutions	Marks	Marker's Comments
(ii)	Let $t = \tan \beta$ $\tan 2\beta = \frac{2t}{1-t^2}$ $\frac{\cos^2 \theta}{2 \sin \theta} = \frac{2t}{1-t^2}$ $\cos^2 \theta - t^2 \cos^2 \theta = 4t \sin \theta$ $0 = t^2 \cos^2 \theta + 4t \sin \theta - \cos^2 \theta$ $t = \frac{-4 \sin \theta \pm \sqrt{16 \sin^2 \theta + 4 \cos^4 \theta}}{2 \cos^2 \theta}$ $= \frac{-4 \sin \theta \pm 2 \sqrt{4 \sin^2 \theta + (1 - \sin^2 \theta)^2}}{2 \cos^2 \theta}$ $= \frac{-4 \sin \theta \pm 2 \sqrt{4 \sin^2 \theta + 1 - 2 \sin^2 \theta + \sin^4 \theta}}{2 \cos^2 \theta}$ $= \frac{-4 \sin \theta \pm 2 \sqrt{1 + 2 \sin^2 \theta + \sin^4 \theta}}{2 \cos^2 \theta}$ $= \frac{-4 \sin \theta \pm 2 \sqrt{(1 + \sin^2 \theta)^2}}{2 \cos^2 \theta}$ $= \frac{-4 \sin \theta \pm 2(1 + \sin^2 \theta)}{2 \cos^2 \theta}$ $= \frac{-4 \sin \theta \pm 2(1 + \sin^2 \theta)}{2 \cos^2 \theta}$ $= \frac{-\sin^2 \theta - 2 \sin \theta - 1}{\cos^2 \theta} \text{ or } \frac{1 - 2 \sin \theta + \sin^2 \theta}{\cos^2 \theta}$ $= -\left[\frac{1 + \sin \theta}{\cos \theta}\right]^2 \text{ or } \left[\frac{1 - \sin \theta}{\cos \theta}\right]^2$ But $0 < \beta < \pi/4$ $\therefore \tan \beta = \left[\frac{1 - \sin \theta}{\cos \theta}\right]^2$	(3)	① Quadratic in $\tan \beta$ ① change of $\cos^4 \theta$ to $(1 - \sin^2 \theta)^2$ ① full solution

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
(iii)  Let angle between OP and asymptote be α. Asymptote: $y=x$ gradient = 1 $\tan \alpha = \frac{\text{grad of asymptote} - \text{mOP}}{1 - \text{grad asymptote} \times \text{mOP}}$ $= \frac{1 - \sin \theta}{1 + 1 \times \sin \theta}$ $= \frac{1 - \sin \theta}{1 + \sin \theta} \times \frac{1 - \sin \theta}{1 - \sin \theta}$ $= \frac{(1 - \sin \theta)^2}{1 - \sin^2 \theta}$ $= \frac{(1 - \sin \theta)^2}{\cos^2 \theta}$ $= \left[\frac{1 - \sin \theta}{\cos \theta} \right]^2$ $= \tan^2 \beta$ as α and β are acute $\therefore \alpha = \beta$ $\therefore$ MP is parallel to asymptote $y=x$ as alternate angles are equal	(3)	① sub into formula. ① showing $\tan \alpha = \tan \beta$ ① explanation